

7 (a) 1. 2.	vibration / oscillation <u>parallel</u> to energy transmission / wave	B1
	vibration / oscillation <u>perpendicular</u> to energy transmission / wave energy transferred	B1
(b)(i)	$7.98(t - 170) = 4.75t$ $t = 420 \text{ s}$ OR $7.98t = 4.75(t + 170)$ $t = 250 \text{ s}$ $250\text{s} + 170\text{s} = 420\text{s}$ $t = \underline{7.0 \text{ min}}$	C1 A1
(ii)	$420 \times 4.75 = 2000 \text{ km}$	B1
(c)(i)	(From Fig 7.3) S-wave <u>velocity is 0/ has no velocity in outer core (and inner core)/ from a depth of 3000km to the center of the Earth</u> S-waves <u>cannot travel through</u> (outer) core Because the (outer) <u>core is liquid/ S-waves cannot travel though liquid</u>	B1 B1 B1
(ii)	These regions are <u>on the other side of the Earth</u> with respect to the epicenter/earthquake	B1
(d)(i)	Speed <u>increases</u> Because (any 2 from): <u>density</u> increases; <u>elastic moduli</u> increases; <u>pressure</u> increases;	M1 B1 B1
(ii)	path curves OR refracted away from centre of earth OR towards surface	M1 A1
(e)(i)	waves that just enter the outer core and those that just scrape/curve across its surface are <u>travelling/separated/ "bent"/"curved" in opposite directions</u> as seen in Fig 6.4. leads to a gap/"region where no seismic waves are detected" between these two positions	B1 B1
(ii) 1. 2.	(From Fig 7.3), Radius of Earth = 6500 km (accept: 6250 km to 6750 km) {do not accept: 6000 km} (From Fig 7.4), $\Delta \theta$ from the centre of the Earth = $143^\circ - 103^\circ$ = 40° = 0.698 rad $s = r(\Delta \theta) = (6250 \text{ km to } 6750 \text{ km})(0.698)$ = 4360 to 4710 km (or 4000 to 5000 km)(1 sf)	B1 C1 A1
(f)	Any one from the following, from any "category": social: locate earthquakes locate landslides warn about tsunamis / volcanoes rush assistance quickly	B1

	identify archaeological sites save lives / evacuate area academic curiosity identify erosion technological: design buildings that resist earthquakes locate gas / oil reserves locate mineral reserves locate water locate sunken treasure / aeroplane Black boxes / oil tanks predict / understand earthquakes / tsunamis / volcanoes economic: cheaper resources no need to replace destroyed buildings discover contamination harness geothermal energy	
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